

claude-windows / 144a31b7-7962-4fee-9d6f-8a7854daa63f

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\144a31b7-7962-4fee-9d6f-8a7854daa63f\subagents\workflows\wf_1c5f1f5c-9ba\agent-a4b7e448820c08c12.jsonl`
- SHA-256: `152ae4d6038ae24834f8b88bf1347c7f9897732539f7e504bf01c96984031b78`
- Source modified: `2026-06-20T08:05:06+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_1c5f1f5c-9ba`
- session_id: `144a31b7-7962-4fee-9d6f-8a7854daa63f`
```

Transcript

1. user (2026-06-20T08:03:26.402Z)

Design the KhelSutra "interactive per-rally selection" feature using this lens:
GRID/LIST-FIRST: the dominant surface is a responsive grid of rally cards (each with a thumbnail/duration/index), with filters (min length, which game, sort) and bulk + individual checkbox selection, plus a sticky "build reel" tray showing the running count/duration.

The feature (owner's words): the engine detects rallies -> present them in a UI so the user PICKS which rallies go into the highlight reel ? dynamic per-video, NOT a fixed threshold. It lives in the alpha SPA (kheLSutra repo, web/) over the rally index the engine produces. It pairs with the conversational "ask for a reel" builder.

GROUND EVERY DECISION in this real research about the actual engine (data contract, API, UI plans, doc convention). Do NOT propose consuming fields that don't exist today (shot type, player, score are roadmap-only unless the research says otherwise). Be honest about what needs new engine API vs what exists:

```
{
  "contract": {
    "tables": [
      {
        "name": "videos",
        "purpose": "One row per ingested video file. Parent of all segments (FK cascade). Status is the ingest verdict (processed/failed). user_id (v4) is NULL on local installs.",
        "columns": [
          "id TEXT PRIMARY KEY (= MD5 of the whole file, compute_video_id)",
          "filepath TEXT NOT NULL",
          "processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP",
          "status TEXT NOT NULL (e.g. 'processed' | 'failed')",
          "validation_results TEXT (JSON list of ValidationResult dicts)",
          "user_id TEXT (NULLABLE, added v4; NULL = local install)"
        ]
      },
      {
        "name": "play_segments",
        "purpose": "CONFIRMED rallies ? the table a per-rally selection UI consumes today. /api/compile and /api/query read here. One row per accepted rally.",
        "columns": [
          "id INTEGER PK AUTOINCREMENT (the segment_id the UI selects/stitches)",
          "video_id TEXT NOT NULL (FK videos.id ON DELETE CASCADE)",
          "start_time REAL NOT NULL (seconds)",
          "end_time REAL NOT NULL (seconds)",
          "duration REAL NOT NULL (= end-start, computed at insert)",
          "server TEXT (NULLABLE; only the motion segmenter sets it, and it is FABRICATED/random)",
          "receiver TEXT (NULLABLE; same ? motion-only, fabricated)",
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        "metadata TEXT (extensible JSON; gemini writes {shot_count, shot_count_confidence,
detector}); motion writes {shot_count, intensity, avg_speed_kmh} all random)"
    ]
},
{
    "name": "candidate_segments",
    "purpose": "RAW pre-confirmation detections (detector-agnostic), persisted by the hybrid
segmenters only. Holds the human-correction-loop columns
(human_label/notes/confirmed_segment_id) that a review UI would write ? but NO endpoint writes
them yet.",
    "columns": [
        "id INTEGER PK AUTOINCREMENT",
        "video_id TEXT NOT NULL (FK videos.id ON DELETE CASCADE)",
        "detector TEXT NOT NULL (e.g. wasb_hybrid/yolo_hybrid/fusion key)",
        "chunk_index INTEGER (NULLABLE)",
        "start_time REAL NOT NULL",
        "end_time REAL NOT NULL",
        "duration REAL NOT NULL",
        "confidence REAL (NULLABLE)",
        "stats TEXT (detector-specific JSON metrics: peak/mean velocity, active_frame_count,
net_crossings, etc.)",
        "detection_params TEXT (JSON of the detector knobs used)",
        "confirmed_segment_id INTEGER (NULLABLE FK play_segments.id ON DELETE SET NULL ? link
to the confirmed rally)",
        "human_label TEXT (NULLABLE ? correction-loop label, UNUSED by any code path today)",
        "notes TEXT (NULLABLE ? correction-loop notes, UNUSED today)",
        "created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP",
        "user_id TEXT (NULLABLE, added v4)"
    ]
},
{
    "name": "events",
    "purpose": "Sub-rally events keyed to a play_segment. ONLY the motion segmenter writes
these, and every value is RANDOM/fabricated (not ground truth). Gemini/hybrid paths write NO
events.",
    "columns": [
        "id INTEGER PK AUTOINCREMENT",
        "segment_id INTEGER NOT NULL (FK play_segments.id ON DELETE CASCADE)",
        "timestamp REAL NOT NULL",
        "type TEXT NOT NULL ('serve' | 'smash' | 'foul' | 'winner')",
        "player TEXT (NULLABLE; from a fabricated player_pool)",
        "details TEXT (JSON; e.g. {speed_kmh}, {type})"
    ]
},
{
    "name": "jobs",
    "purpose": "Async job store (v2/v3). One row per /api/process submission. status =
EXECUTION state; the pipeline verdict lives inside result JSON. Not rally data, but the UI polls
GET /api/jobs/{id} for progress.",
    "columns": [
        "id TEXT PK (uuid hex)",
        "video_path TEXT NOT NULL",
        "video_id TEXT (NULLABLE)",
        "segmenter TEXT",
        "player_pool TEXT (JSON)",
        "max_frames INTEGER",
        "status TEXT NOT NULL DEFAULT 'queued' (queued|running|waiting|done|failed)",
        "progress TEXT",
        "error TEXT",
        "result TEXT (JSON full response dict incl. play_segments + report paths)",
        "created_at/started_at/finished_at TIMESTAMP",
        "policy TEXT (v3 JSON ExecutionPolicy)",
        "next_poll_at TIMESTAMP (v3)"
    ]
},
},

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{
  "name": "user_models",
  "purpose": "Per-user model store (v5). Personalization groundwork ONLY ? nothing on the
served path reads it. Not consumed by a rally-selection UI.",
  "columns": [
    "id, user_id, version, baseline_version, feature_schema_hash, fusion_config(JSON),
classifier_json(JSON), promotion_evidence(JSON), n_videos_at_fit, is_active, created_at"
  ]
},
],
"rally_fields_today": [
  "play_segments.id ? the integer the UI selects and passes to /api/compile as segment_ids
(backend/infrastructure/database.py:125-135, backend/main.py:102-105,386)",
  "start_time / end_time / duration (REAL seconds) ? the interval the reel is cut from;
duration computed at insert as end-start (database.py:127-129,347)",
  "video_id ? links the rally to its source file; /api/compile resolves videos.filepath as
the raw source to cut (main.py:344,393; database.py:418-425)",
  "metadata JSON.shot_count + shot_count_confidence ? gemini (default) persists these from
Gemini's shuttle_exchange_count + confidence; falls back to max(3,int(dur/0.8)) when absent
(gemini.py:470-485)",
  "metadata JSON.detector ? the model_name string (e.g. 'gemini-2.5-flash')
(gemini.py:484)",
  "server / receiver ? present in schema and returned by query, but ONLY the 'motion'
segmenter sets them and they are random.sample fabrications, NOT real player identity
(motion.py:191,201; database.py:130-131)",
  "events (type/player/timestamp/details) ? serve/smash/foul/winner; ONLY motion writes them
and ALL values are random.* fabrications, queryable via /api/query event_type filter
(motion.py:177-252; database.py:451-453)",
  "candidate_segments.stats JSON (hybrid paths only) ? per-window peak/mean velocity,
active_frame_count, net_crossings; raw detector metrics, not surfaced by /api/query
(database.py:362-383)",
  "candidate_segments.confidence ? REAL, persisted by add_candidate_segment but currently
passed as None by the trajectory hybrids (database.py:163,362-383)"
],
"rally_fields_roadmap": [
  "shot_type (smash/clear/drop/drive/net) per shot ? NO detection or column exists;
events.type only carries motion's fabricated serve/smash/foul/winner labels, never produced by
gemini/hybrid paths (motion.py:228-239)",
  "player identity per rally/shot ? NO real detection. play_segments.server/receiver and
events.player are random.sample picks from a hardcoded pool in the motion demo only;
PersonDetector tracks anonymous chunk-local ByteTrack IDs, never named players
(motion.py:25-26,191; person_detector.py make_tracker)",
  "match score / point score per rally ? ABSENT everywhere. No score/game/set column in any
table; grep for score/game_index/set_index in backend finds only eval-F1 'score' and a
detector_score_threshold config knob, never a match score (database.py whole file;
config/models.py:126)",
  "game/set index (which game of the match a rally belongs to) ? NO column and no detection;
rallies are a flat per-video list with no game/set grouping (database.py:122-135)",
  "confidence on confirmed rallies ? play_segments has no confidence column; only
candidate_segments.confidence exists and the live trajectory hybrids pass None into it
(database.py:122-135 vs 362-383)",
  "human correction state (accepted/rejected/nudged) ? columns EXIST
(candidate_segments.human_label, notes, confirmed_segment_id) but are dead: no API endpoint or
code path writes human_label/notes; UI_PROPOSAL B6 PATCH /api/segments/{id} is proposed, not
built (database.py:166-168; docs/UI_PROPOSAL.md:59)"
],
"reel_compose": {
  "entrypoint": "POST /api/compile -> backend/main.py:342-396 (compile_highlights) ->
backend/pipeline/stitcher/compiler.py:303 VideoCompiler.compile_highlights. Batch/CLI variants:
tools/generate_highlights.py and main.py --trim-start/--trim-end both call the same
VideoCompiler.",
  "inputs": "CompileRequest = {video_id: str, segment_ids: List[int], output_filename: str}
(main.py:102-105). The endpoint loads videos.filepath via db.get_video, fetches ALL
play_segments via query_play_segments, filters to the requested segment_ids, applies the
stitching.min_shot_count floor (segments without a known shot_count are exempt), then extracts

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time_pairs = [(start_time,end_time)] (main.py:344-386).
VideoCompiler.compile_highlights(raw_video_path, segments:[(start,end)], output_filename) is the
underlying contract ? it takes the RAW source path + interval pairs, NOT segment ids
(compiler.py:303).",
    "outputs": "A single MP4 written to output_dir/<safe output_filename>; the endpoint
returns {status:'success', filepath: out_path} (main.py:393-394). The browser then downloads it
via GET /api/highlights/{video_id}/{filename} (FileResponse stream; main.py:307-328). There is
NO summary/JSON sidecar emitted by compile ? the per-video observability report is a separate
artifact emitted during processing, not at compile time.",
    "notes": "Pipeline: _merge_overlapping (sort + collapse overlapping/abutting/duplicate
ranges so a rally never plays twice; drops end<=start) -> per-clip ffmpeg trim with
begin_padding/end_padding (default 0.5/0.5 from StitchingConfig) RE-ENCODED libx264 superfast
crf22 + aac, optional drawtext/logo watermark baked per-clip -> concat demuxer with -c copy
(clips are codec-identical so concat stream-copies). Out-of-range/invalid segments are skipped
with a logged summary; if ALL skip it raises. Selection is purely by segment_ids the caller
supplies ? the backend has no 'select best N' logic."
    },
    "playback": {
        "proxy_available": "NO. There is no low-res proxy video generated or stored for UI
scrubbing. The only downscaling is transient: gemini.py re-encodes/downscales upload chunks to
indexing.ai_chunk_scale_width (default 1280) purely to fit the provider upload cap, and those
chunk files are deleted immediately after analysis (gemini.py:189-235,531). No proxy/preview MP4
is persisted anywhere (grep for proxy/preview/low-res finds only docs + the chunk-scale knob).",
        "frame_access": "NO per-rally thumbnails or extracted frames exist for the UI. The only
frame extraction is wasb_infer.py::extract_frames
(backend/pipeline/detectors/wasb_infer.py:547-571) which decodes the whole video to index-named
PNGs (00000000.png ...) inside the WSL detector cache as a transient input to the GPU shuttle
detector; it is deleted on a successful run by the cache-hygiene logic (keep_frames default
false) and is never keyed to rallies or served. There is no thumbnail column and no /api/frames
endpoint.",
        "notes": "A per-rally clip UI today has two options, both requiring the RAW source file:
(1) play the single compiled reel after /api/compile + download via GET
/api/highlights/{video_id}/{filename}; (2) play the full source and seek to
play_segments.start_time/end_time (the times are exact seconds). There is NO endpoint that
streams an arbitrary [start,end] span or a single rally clip, and NO endpoint that serves the
original source video ? only compiled reels under output_dir are downloadable (main.py:307-328).
The current static UI (backend/api/static/app.js) has NO <video> element at all: it renders
rally cards, lets the user check segment_ids, compiles, and offers a download link; it also
reads data.rallies which is a known bug (the API returns play_segments), per UI_PROPOSAL B5
(app.js:344,378; UI_PROPOSAL.md:58). A real review UI would need to either serve the source file
or add a per-span clip/proxy endpoint ? neither exists."
    },
    "file_refs": [
        "backend/infrastructure/database.py:109-148 (base tables: videos, play_segments, events)",
        "backend/infrastructure/database.py:122-135 (play_segments columns)",
        "backend/infrastructure/database.py:137-148 (events columns)",
        "backend/infrastructure/database.py:150-178 (v1 candidate_segments + columns)",
        "backend/infrastructure/database.py:166-168 (human_label, notes, confirmed_segment_id
correction-loop columns)",
        "backend/infrastructure/database.py:180-205 (v2 jobs table)",
        "backend/infrastructure/database.py:207-217 (v3 jobs.policy/next_poll_at)",
        "backend/infrastructure/database.py:219-237 (v4 user_id on videos + candidate_segments)",
        "backend/infrastructure/database.py:239-285 (v5 user_models)",
        "backend/infrastructure/database.py:341-353 (add_play_segment: duration = end-start)",
        "backend/infrastructure/database.py:362-383 (add_candidate_segment, confidence/stats)",
        "backend/infrastructure/database.py:427-475 (query_play_segments + filters + events join ?
the UI read path)",
        "backend/pipeline/segmenters/gemini.py:461-498 (_populate_db: metadata = shot_count,
shot_count_confidence, detector; server/receiver = None)",
        "backend/sports/badminton.py:41-54 (Gemini output schema:
start_time,end_time,shuttle_exchange_count,shot_count_confidence ? no shot_type/player/score)",
        "backend/pipeline/segmenters/motion.py:191-250 (fabricated server/receiver + random
serve/smash/foul/winner events)",
        "backend/main.py:102-105 (CompileRequest = video_id, segment_ids, output_filename)",
        "backend/main.py:342-396 (/api/compile: select by segment_ids, min_shot_count floor,

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time_pairs, VideoCompiler)",
    "backend/main.py:307-328 (/api/highlights/{video_id}/{filename} download ? only compiled
reels, not source)",
    "backend/main.py:331-340 (/api/query returns play_segments)",
    "backend/pipeline/stitcher/compiler.py:80-105 (_merge_overlapping de-dupe)",
    "backend/pipeline/stitcher/compiler.py:303-413 (compile_highlights: per-clip re-encode ->
concat -c copy -> single MP4)",
    "backend/pipeline/detectors/wasb_infer.py:547-571 (extract_frames: transient WSL PNG
frames, deleted on success ? not UI thumbnails)",
    "backend/api/static/app.js:344,363-409 (UI: no video element, data.rallies bug, select
segment_ids -> compile -> download)",
    "backend/config/models.py:126 (detector_score_threshold ? a config knob, NOT a per-rally
match score)",
    "docs/UI_PROPOSAL.md:46,58-59 (A4 rally-review + B5 replace static UI + B6 PATCH
/api/segments correction endpoints ? proposed, NOT implemented)"
]
},
"api": {
    "endpoints": [
        {
            "method": "GET",
            "path": "/",
            "purpose": "Serve the bundled static SPA shell (backend/api/static/index.html) or a
placeholder message.",
            "response": "HTML (HTMLResponse). text/html string.",
            "auth": "None. backend/main.py:52"
        },
        {
            "method": "GET",
            "path": "/api/config",
            "purpose": "Return the live typed AppConfig the server started with (Pydantic model
dump).",
            "response": "JSON: the full AppConfig object (indexing/stitching/output/security/etc).
backend/main.py:121-123 returns global_config.",
            "auth": "None. Leaks full server config incl. security settings. backend/main.py:121"
        },
        {
            "method": "POST",
            "path": "/api/process",
            "purpose": "Submit a hash->validate->segment job for a server-local video_path. Fast-
fails bad path / missing file / unknown segmenter, then enqueues async job.",
            "response": "202 JSON {job_id, status:'queued', poll:'/api/jobs/{job_id}'}. Body model
ProcessRequest{video_path, player_pool?, max_frames?, segmenter?}. backend/main.py:240-271",
            "auth": "None. Path containment only IF security.allowed_video_root is set
(validate_input_path), else any server-readable path. backend/main.py:240"
        },
        {
            "method": "GET",
            "path": "/api/jobs/{job_id}",
            "purpose": "Poll execution state of a submitted processing job.",
            "response": "JSON {job_id, status(queued|running|waiting|done|failed), progress,
video_id, error, result(pipeline dict incl. play_segments), reports, created_at, started_at,
finished_at}. 404 if unknown. backend/main.py:274-293",
            "auth": "None. Any caller can poll any job_id (no ownership check). backend/main.py:274"
        },
        {
            "method": "POST",
            "path": "/api/query",
            "purpose": "List/filter play (rally) segments for a video_id. This is the ONLY way to
list rallies today.",
            "response": "JSON {play_segments: [ {id, video_id, start_time, end_time, duration,
server, receiver, metadata(dict), events:[{id,segment_id,timestamp,type,player,details}]} ]}.
Body QueryRequest{video_id, server?, receiver?, duration_min?, event_type?}.
backend/main.py:331-340; shape from database.py:427-475",
            "auth": "None. Any caller can query any video_id's rallies. backend/main.py:331"
        }
    ]
}

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    },
    {
        "method": "POST",
        "path": "/api/compile",
        "purpose": "Stitch a CUSTOM set of selected rally segment_ids into one highlight reel
(applies stitching.min_shot_count floor + paddings + watermark).",
        "response": "JSON {status:'success', filepath: server-local out path}. 404 unknown
video; 400 no matching/all-below-floor; 500 stitch fail. Body CompileRequest{video_id,
segment_ids:[int], output_filename}. backend/main.py:342-396",
        "auth": "None. Returns a server-local filepath (not the download URL).
backend/main.py:342"
    },
    {
        "method": "GET",
        "path": "/api/highlights/{video_id}/{filename}",
        "purpose": "Stream/download a previously compiled highlight reel by bare filename.",
        "response": "Binary FileResponse (video/mp4 or video/quicktime). 400 bad/traversal name;
404 unknown video or no compiled file. backend/main.py:307-328",
        "auth": "None. Path-traversal guarded (safe_output_filename + resolve_under_root).
backend/main.py:307"
    },
    {
        "method": "POST",
        "path": "/api/upload/init",
        "purpose": "Begin a chunked browser upload (validates extension + total-size cap).",
        "response": "JSON {upload_id, max_part_mb, next_part:'/api/upload/{id}/part/0'}. 400 bad
ext/name; 413 too big. Body UploadInitRequest{filename, total_size_bytes?}.
backend/api/uploads.py:64-86",
        "auth": "None. Upload state is in-process, global (not per-user).
backend/api/uploads.py:64"
    },
    {
        "method": "PUT",
        "path": "/api/upload/{upload_id}/part/{index}",
        "purpose": "Append one strictly-sequential raw-bytes part to an in-flight upload
(enforces per-part + total caps).",
        "response": "JSON {upload_id, received_parts, received_bytes}. 404 unknown id; 409 out-
of-order; 413 overflow (aborts). backend/api/uploads.py:89-120",
        "auth": "None. backend/api/uploads.py:89"
    },
    {
        "method": "POST",
        "path": "/api/upload/{upload_id}/complete",
        "purpose": "Finalize a chunked upload: verify part count, move into upload_dir, compute
video_id.",
        "response": "JSON {path(abs server path), video_id(MD5), size_bytes, next:'POST
/api/process with this path'}. 404 unknown id; 409 part-count mismatch. Body
UploadCompleteRequest{total_parts}. backend/api/uploads.py:123-144",
        "auth": "None. backend/api/uploads.py:123"
    },
    {
        "method": "DELETE",
        "path": "/api/upload/{upload_id}",
        "purpose": "Abort an in-flight upload and delete its partial file.",
        "response": "JSON {status:'aborted', upload_id}. 404 unknown id.
backend/api/uploads.py:147-155",
        "auth": "None. backend/api/uploads.py:147"
    }
},

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"auth_model": "NONE today. Every endpoint is fully unauthenticated: there is no Depends/Security dependency, no HTTPBearer, no API-key/token check, no DEV_USER_EMAIL read, and no per-user scoping in any handler (verified across backend/**/.py). CORS is wide-open: allow_origins=['*'] with allow_credentials=False (backend/main.py:40-46). The DB schema is auth-READY but unwired: videos.user_id and candidate_segments.user_id columns + partial indexes exist (database.py:221-236, migration v4) and a per-user user_models table exists

(database.py:240-283, v5), but NO endpoint reads or writes user_id. INTENDED model (docs, not built): alpha = Cloudflare Access in front of both hostnames; FastAPI trusts the 'Cf-Access-Authenticated-User-Email' header (verify the Access JWT) for per-user scoping (HOSTING_PLAN.md:37-38). Local-dev fallback DEV_USER_EMAIL env + an owner_email column + per-query scoping + narrowed CORS + rate limit + per-user quota are all scoped to the UNSTARTED PR-3 (UI_ALPHA_EXECUTION_PLAN.md:32). Beta graduates to Firebase Auth (HOSTING_PLAN.md:62-63). Net: ship-blocking gap for any multi-tenant/hosted use; fine only for the single-box local alpha it currently is.",

"gaps": [

"NO 'list videos / list matches' endpoint at all. There is no GET /api/videos and no DB method to enumerate videos (no get_all_videos/list_videos in database.py; only get_video(video_id) at database.py:418 which requires you to ALREADY know the id). An interactive UI cannot populate a match list without first knowing every video_id out-of-band. UI_PROPOSAL screen A1 'Matches' has no backing endpoint.",

"NO GET endpoint that returns rallies/segments for a video as JSON. The only way to list rallies is POST /api/query (backend/main.py:331), which is a body-based RPC, not a RESTful GET /api/videos/{video_id}/rallies. Functionally usable for per-rally selection (it returns id, start_time, end_time, duration, metadata, events), but not cacheable/linkable and shaped as a filter-query, not a resource listing.",

"NO per-rally thumbnail or frame endpoint. There is no GET /api/videos/{video_id}/rallies/{id}/thumbnail and no frame-serving route (grep for /thumbnail|/frame|/proxy = none). A per-rally-selection grid UI cannot show preview stills; no frame-extraction handler exists.",

"NO endpoint to serve/stream the SOURCE video or a low-res proxy for scrubbing. Only the COMPILED reel is streamable (GET /api/highlights/{video_id}/{filename}, backend/main.py:307). A rally-review UI that lets a user seek/preview each rally in the original footage has nothing to load (UI_ALPHA_EXECUTION_PLAN M3 'player seek' is unbacked).",

"Custom-selection -> reel EXISTS (POST /api/compile with arbitrary segment_ids, backend/main.py:342), but it does NOT return a download URL ? it returns a server-local filepath ('filepath': out_path, backend/main.py:394). The client must separately construct GET /api/highlights/{video_id}/{output_filename}. A cleaner API would return the ready download URL.",

"NO correction / accept-reject / boundary-nudge endpoint. There is no PATCH /api/segments/{id}; the candidate_segments.human_label/notes columns exist but no route writes them. This is explicitly the unstarted PR-4 'B6 correction loop' (UI_ALPHA_EXECUTION_PLAN.md:33). An interactive per-rally UI cannot persist user corrections today.",

"NO endpoint links a completed job to a video for follow-up. GET /api/jobs/{job_id} (backend/main.py:274) returns video_id only while the job row exists; there is no GET /api/videos/{video_id}/status to re-discover processing state later, and no listing of a video's compiled reels.",

"NO /api/health endpoint, despite HOSTING_PLAN.md:42 expecting the SPA to ping one for the 'engine offline' banner. Not built.",

"Auth/ownership gap for the UI: with no auth and no user scoping, the API cannot show a per-user match list or prevent one tester from reading/compiling another's rallies (videos.user_id exists at database.py:221-236 but is never set or filtered on). PR-3 identity scoping is unstarted (UI_ALPHA_EXECUTION_PLAN.md:32)."

],

"file_refs": [

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\main.py",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\api\\uploads.py",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\api\\job_worker.py",

"C:\\Users\\avidu\\Projects\\badminton-highlight-

indexer\\backend\\infrastructure\\database.py",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\api\\static\\app.js",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\config\\models.py",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\HOSTING_PLAN.md",

"C:\\Users\\avidu\\Projects\\badminton-highlight-

indexer\\docs\\UI_ALPHA_EXECUTION_PLAN.md",

"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md"

]

},

"ui": {

"spa_plan": {

"stack": "React + Vite + TypeScript + Tailwind SPA, single-page, client-side router, typed

API client; deployed as static hosting (Cloudflare Pages, \$0) talking to the existing FastAPI backend on the owner's GPU box via a Cloudflare Tunnel at api.khelsutra.guru. Auth in alpha = Cloudflare Access email-allowlist (OTP), zero auth code; backend reads the Cf-Access-Authenticated-User-Email header for per-user scoping. CLI is NOT replaced ? SPA hits the same FastAPI surface. Stack 'stands unless owner objects' (still listed open in execution plan). SPA lives in a SEPARATE private repo `khelsutra` under web/; backend enabler PRs stay in the indexer repo. Refs: UI_PROPOSAL.md:33-37, UI_ALPHA_EXECUTION_PLAN.md:13-15;35-40, HOSTING_PLAN.md:20-34.",

"milestones": [
"SPA milestones (khelsutra repo web/, UI_ALPHA_EXECUTION_PLAN.md:35-40): M1 scaffold ? Vite+React+TS+Tailwind, router, typed API client, KhelSutra Guru branding.",
"M2 ? A2 Upload (chunked + recording-guidelines checklist) + A3 Job Progress (poll).",
"M3 ? A1 Matches + A4 Rally Review (player seek + accept/reject/nudge wired to backend PR-4).",
"M4 ? A5 Highlights (compile + download) + A6 Report + engine-offline banner; plus unit tests for API client + review-state logic and a clean `npm run build`.",
"Backend enabler PRs gating the SPA (L1, indexer repo): PR-1 B1 async job model #16 (MERGED #87); PR-2 B2+B3 chunked upload + streaming download (MERGED #99); PR-3 B4+B7 identity & exposure (owner_email scoping, CORS, rate limit, quota) ? open; PR-4 B6 correction loop (PATCH /api/segments/{id}) + retire static UI (B5) ? open.",
"Phase structure (UI_ALPHA_EXECUTION_PLAN.md:19-55): G0 setup ? Phase L (local build+test: L1 backend, L2 SPA M1-M4, L3 local E2E, GATE GL owner sign-off) ? Phase D (deployment D1-D3, GATE GD) ? Phase R (rollout wave 1: 5 testers ? widen to 25). Beta is a separate 4-Drop roadmap (UI_PROPOSAL.md:73-93)."

],
"routes": [
"A1 Matches (home) ? user's videos, per-video status/verdict chip, upload CTA (UI_PROPOSAL.md:43)",
"A2 Upload & Process ? drag-drop, inline recording-guidelines checklist, chunked upload, auto-start processing (UI_PROPOSAL.md:44)",
"A3 Job Progress ? stage list (validation?segmentation?AI confirm?report) with live poll of GET /api/jobs/{id}, verdict banner (UI_PROPOSAL.md:45)",
"A4 Rally Review (flagged ?, 'the corpus engine') ? player + rally list, click-to-see, per-rally Accept/Reject/nudge $\pm 0.5s$, bulk 'all good' (UI_PROPOSAL.md:46)",
"A5 Highlights ? select accepted rallies ? name reel ? compile ? download + list of past reels (UI_PROPOSAL.md:47)",
"A6 Report ? customer-audience report (verdict, notes, FAQ refs) + 'nerd view' toggle for technical.md (UI_PROPOSAL.md:48)"

],
"components": [
"6-screen alpha set A1-A6 (UI_PROPOSAL.md:39-48).",
"Rally-review widget: video player + rally list with click-to-see, accept/reject toggles, start/end nudge control ($\pm 0.5s$), bulk-accept, per-rally chips (duration, shot count + confidence) ? backed by B6 PATCH /api/segments/{id} ? candidate_segments.human_label/notes + provenance (UI_PROPOSAL.md:46;59, UI_ALPHA_EXECUTION_PLAN.md:33;38).",
"Chunked uploader (?90-95 MB parts, mp4/mov allowlist, size cap) + recording-guidelines checklist for pre-upload education (UI_PROPOSAL.md:44;55, HOSTING_PLAN.md:38-39).",
"Job-progress poller against GET /api/jobs/{id} (UI_PROPOSAL.md:45).",
"Highlights compiler/download via /api/compile + streaming GET /api/highlights/{video_id}/{name} (UI_PROPOSAL.md:47;56).",
"Report renderer with nerd-view toggle (UI_PROPOSAL.md:48).",
"Typed API client + review-state logic (unit-tested) (UI_ALPHA_EXECUTION_PLAN.md:36;40).",
"Engine-offline banner driven by /api/health ping (UI_ALPHA_EXECUTION_PLAN.md:39, HOSTING_PLAN.md:43)."

],
"conversational_design": "IMPORTANT / honest finding: NONE of the four docs read (UI_PROPOSAL, UI_ALPHA_EXECUTION_PLAN, HOSTING_PLAN, KHELSTRA_PRODUCT_OVERVIEW) specify a conversational 'ask for a reel' query feature, and there is NO mapping of a plain-English request to TODAY's metadata (length / count / order / which game) anywhere in them. The alpha Highlights flow (A5) is explicitly a MANUAL select-then-compile: user selects accepted rallies ? names the reel ? compiles ? downloads (UI_PROPOSAL.md:47). Today's metadata surfaced to the user is per-rally duration + shot count (+confidence) as chips on the Rally Review screen (UI_PROPOSAL.md:46) and a summary of rally count + total play time

(KHELSUTRA_PRODUCT_OVERVIEW.md:14;48;65). 'Length and order' appears ONLY as a FEEDBACK prompt to alpha testers ('Whether the reel is the right length and order for how you'd actually use it', KHELSUTRA_PRODUCT_OVERVIEW.md:135) ? i.e. a question asked OF the user, not a query the user can issue. Conversational QA / NL query is EXPLICITLY DEFERRED: 'no conversational QA (deferred in roadmap)' (UI_PROPOSAL.md:66). Where a conversational layer IS specified (outside these four docs, all future/gated): OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55 ('build the bookkeeping NOW, punt the model' ? punt the conversational feature until extraction quality is solid) with the queryable per-video event-index schema as 'the conversational seam' (lines 102, 160); PLATFORM_ARCHITECTURE.md:263-272 (NL queries about rallies/shots/technique, must be localized); I18N_PLAN.md:159 (P4 'Conversational + coach-mode (future): locale-aware NL query intake ? templated localized answers ? gated on those features existing'). So the metadata that COULD back a future query (rally_ordinal, start/end ? length, shot_count, which-game/match keying) is being reserved in the event-index schema (POST_166?MILESTONES.md:55), but no length/count/order/which-game parsing of a plain-English reel request is designed in the read docs. Explicitly deferred dimensions: shot-type and player are both roadmap-only ? shot detection (smashes/drops/net/serves/clears) is 'on the roadmap, not available today' (KHELSUTRA_PRODUCT_OVERVIEW.md:116;160) and player tracking/identification is a hard non-goal / strictly-future biometrics (KHELSUTRA_PRODUCT_OVERVIEW.md:160;181, and the alpha non-goal 'no athlete profiles', UI_PROPOSAL.md:64).",

"positioning_notes": [

"Per-rally selection IS in the alpha product, but as TWO distinct surfaces: (a) Rally Review (A4) for accept/reject/boundary-nudge correction, and (b) Highlights (A5) for picking which accepted rallies go into a named reel that is compiled + downloaded. Both are manual UI interactions, not a query/builder (UI_PROPOSAL.md:46-47).",

"A 'conversational builder' is NOT positioned as a current product feature in any of the four docs. The reel-building experience shipped in alpha is manual select-then-compile (UI_PROPOSAL.md:47); conversational QA is listed as an alpha non-goal, 'deferred in roadmap' (UI_PROPOSAL.md:66).",

"The strategic framing of per-rally selection is the CORPUS-GROWTH ENGINE, not a consumer convenience: A4 Rally Review is flagged ? as 'the corpus engine' ? every accept/reject/nudge produces the labels the owned-model track is starved for, persisted to candidate_segments.human_label/notes with provenance (UI_PROPOSAL.md:9-14;46;59). This realizes the v1 'correction loop' / 'co-builders' positioning (KHELSUTRA_PRODUCT_OVERVIEW.md:92-94).",

"A future conversational surface is positioned as a downstream OWNED-MODEL / platform capability, deliberately decoupled: extract-once/query-many event index is the 'conversational seam' to be built early while the conversational *model/feature* is punted until extraction quality is high (OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55;102;160; PLATFORM_ARCHITECTURE.md:263-272). The product overview frames this as the path 'toward an AI assistant coach' ? direction, not today's buttons (KHELSUTRA_PRODUCT_OVERVIEW.md:111-126).",

"Honesty discipline is explicit in positioning: the overview twice tells users the 'gets sharper over time' / deeper-analysis story is the direction being built toward, NOT a switch already on (KHELSUTRA_PRODUCT_OVERVIEW.md:105-107;124-126)."

],

"available_today": [

"Upload a full session/match video (phone-on-tripod or GoPro; long videos fine), chunked upload (KHELSUTRA_PRODUCT_OVERVIEW.md:44;47, UI_PROPOSAL.md:44)",

"Automatic rally detection ? finds rallies + their start/end (KHELSUTRA_PRODUCT_OVERVIEW.md:45)",

"Highlight reel: rallies stitched into one downloadable/shareable MP4 with a 'Generated by Khelsutra.guru' mark (KHELSUTRA_PRODUCT_OVERVIEW.md:46)",

"Simple summary: rally count + total actual play time (KHELSUTRA_PRODUCT_OVERVIEW.md:14;48)",

"Re-use & download: process once, return later, download the reel; re-processing REPLACES the prior result (KHELSUTRA_PRODUCT_OVERVIEW.md:49;66)",

"Per-rally review + correction: click-to-see, accept/reject, boundary nudge $\pm 0.5s$, bulk-accept, with per-rally duration + shot-count(+confidence) chips (UI_PROPOSAL.md:46) ? the core human correction loop",

"Manual reel building: select accepted rallies ? name ? compile ? download (UI_PROPOSAL.md:47)",

"Customer-audience report with verdict/notes/FAQ + nerd-view toggle (UI_PROPOSAL.md:48)",

"Background/async processing ? close the tab, come back later (KHELSUTRA_PRODUCT_OVERVIEW.md:62-64; async job model PR-1 #87 merged)",

"Backend reality: alpha serves the Gemini segmenter path (no GPU in request path), ~ 0.93 F1 on clean single-court footage, $\sim \$0.05$ /run (UI_PROPOSAL.md:16-18, HOSTING_PLAN.md:32-33)",

"Invite-only, English-only private web app (KHELSUTRA_PRODUCT_OVERVIEW.md:59;156)"

```

],
"roadmap": [
  "Shot detection ? smashes, drops, net shots, serves, clears within each rally
(KHELSUTRA_PRODUCT_OVERVIEW.md:116;160) ? DEFERRED, not today",
  "Deeper per-rally / per-match analysis: patterns, tendencies
(KHELSUTRA_PRODUCT_OVERVIEW.md:117)",
  "Per-match history / progress over weeks-months (KHELSUTRA_PRODUCT_OVERVIEW.md:118)",
  "AI assistant coach ? coaching insights that support, never replace, the coach
(KHELSUTRA_PRODUCT_OVERVIEW.md:119-120)",
  "Conversational / NL query (the 'ask for a reel' surface): seam/event-index built now,
model/feature punted until extraction quality is solid; must be localized
(OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55, PLATFORM_ARCHITECTURE.md:263-272, I18N_PLAN.md:159
P4) ? DEFERRED (UI_PROPOSAL.md:66)",
  "More sports: table tennis next, then other racket/court sports
(KHELSUTRA_PRODUCT_OVERVIEW.md:121; alpha is badminton-only, UI_PROPOSAL.md:65)",
  "Localization: Kannada + Hindi alongside English (KHELSUTRA_PRODUCT_OVERVIEW.md:122)",
  "Beta drops (UI_PROPOSAL.md:73-93): Drop 1 accounts/academies + real auth (Firebase/Clerk,
18+ only, roles); Drop 2 cloud storage + GPU box as pull-based byo_worker + direct-to-cloud
resumable uploads + email-when-ready; Drop 3 Review v2 (J/K/L + [/]) shortcuts, frame-step,
timeline minimap, batch accept, tags, expiring share links, per-player aggregates ? NO
gait/biometrics); Drop 4 monetization (Razorpay per-match credits) + training-data opt-in
(default OFF, adult-only, minors excluded)",
  "Per-person tracking / biometrics: explicitly NOT today and gated as strictly-future
(KHELSUTRA_PRODUCT_OVERVIEW.md:107;160;181)",
  "Pricing not set (KHELSUTRA_PRODUCT_OVERVIEW.md:162)"
],
"file_refs": [
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_PROPOSAL.md (alpha 6
screens A1-A6 :39-48; A4 'corpus engine' :46; A5 manual compile :47; conversational QA deferred
:66; stack :33-37; beta drops :73-93; backend B1-B7 :52-60)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\UI_ALPHA_EXECUTION_PLAN.md
(SPA milestones M1-M4 :35-40; backend PRs L1 :29-33; phases/gates :19-55; stack-open :15)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\HOSTING_PLAN.md (alpha
topology + Gemini path :20-34; auth header :37-38; 100MB chunk cap :38-39; offline banner :43)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\KHELSUTRA_PRODUCT_OVERVIEW.md (today vs roadmap :40-49 / :111-126; 'right length
and order' feedback prompt :135; shot/player non-goals :160;181; honesty caveats
:105-107;124-126)",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55;102;160 (conversational seam built now,
feature punted) ? NOT among the four docs read but the authoritative location of conversational
design",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\PLATFORM_ARCHITECTURE.md:263-272 (NL query surface, localized) ? supporting",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\I18N_PLAN.md:159 (P4
conversational NL intake, gated) ? supporting",
  "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:55 (event-index schema
fields rally_ordinal, start/end, shot_count) ? the metadata a future query would map onto"
]
},
"convention": {
  "required_sections": [
    "0. TL;DR ? 2-5 sentences: goal, approach, single most important caveat; call out any
standalone-actionable finding (template:30)",
    "1. Problem & goal ? why now; concrete user-facing outcome; what 'good' looks like; link
the code/docs a reader must read to get current (template:33). NOTE: real examples often split
this into '1. Problem & motivating example' + '2. Goal & non-goals'
(PROMINENT_COURT_DETECTION.md:14,19)",
    "2. Decisions locked ? table capturing answered questions so they aren't relitigated
(template:35)",
    "3. Foundation ? research / prior art *(OPTIONAL ? include only when the plan rests on
external facts: cited research, legal/compliance, benchmarks, prior-art; mark researched-vs-
verified; list sign-off gates)* (template:42)",
    "4. Design / architecture ? the actual plan: data contracts, diagrams, and a REUSE MAP

```

naming existing modules/code this builds on so new-vs-reused is clear (template:45)",

"5. Threat model / risk table *(OPTIONAL ? security/privacy/compliance work; who/what is protected vs NOT; residual risks stated plainly)* (template:48). In non-security docs this slot is used for a 'Risks & limits' or 'Experiment/measurement plan' section instead (PROMINENT_COURT_DETECTION.md:77,86)",

"6. Honest limits ? what this does NOT do ? the false-confidence guardrail: what a green suite / DONE status does NOT prove (template:51)",

"7. Deliverables & progress tracker ? SOURCE OF TRUTH ? one small PR per row, branched from origin/master, no stacking; update row status+PR as work lands (template:54)",

"8. Open questions ? owner / external ? split blocking-gates from nice-to-knows; name who decides (template:61)",

"9. Definition of done ? explicit acceptance checklist; when all ticked flip Status->DONE and archive (template:64)",

"10. References ? Internal code anchors [verified] (paths) · Internal docs · External (cited sources) (template:68)",

"Changelog (trailing, after a --- rule) ? dated bullets of what changed (template:73)"

],

"header_fields": [

"Status: a backtick-quoted lifecycle state (DRAFT | IN PROGRESS | DONE) optionally qualified, e.g. `DRAFT` ? `owner-gated for any default flip` or `IN PROGRESS` (template:21, PROMINENT_COURT_DETECTION.md:3, REQUEST_ACCESS_FORM.md:3)",

"Owner: name, e.g. Avi (template:21, PROMINENT_COURT_DETECTION.md:3)",

"Created: YYYY-MM-DD (template:21)",

"Last updated: YYYY-MM-DD (template:21)",

"Lifecycle: literal `DRAFT ? IN PROGRESS ? DONE ? archived` with the note `(move to docs/archives/ when DONE)` (template:22)",

"Tracking anchors: '\$N progress tracker is the source of truth; indexed in docs/README.md; pointer in memory current-state (and docs/NEXT_STEPS.md once work starts)' (template:23, PROMINENT_COURT_DETECTION.md:5)",

"Relation to existing docs: <extends | supersedes | peer-of ?> ? which sibling docs/gaps this attacks (template:24, PROMINENT_COURT_DETECTION.md:6)",

"Honesty note: 'mark each claim [verified] / [researched] / [design]. Not legal/financial advice where applicable.' (template:25, PROMINENT_COURT_DETECTION.md:7)"

],

"progress_tracker_format": "Section §7 titled 'Deliverables & progress tracker' marked '? source of truth' (template:54). It opens with a Legend line then a markdown table. CANONICAL legend (template:55): 'Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row, branched from origin/master, no stacking. Update the row (status + PR link) as work lands.' CANONICAL columns (template:57-59): | ID | Deliverable | Depends on | Gated? | Status | PR | ? first row example: | P0 | ? | ? | No | ? | ? |. IDs are short prefixed codes (P0/P1?, C0/C1?, or tier-coded E1/W1/M1/R1). The Status cell carries the glyph PLUS a short measured result, e.g. '? baseline F1 0.300, merge_rate 0.523' or '? golden ?F1 +0.006 (n.s.)' (RALLY_QUALITY_RESEARCH.md:537-539). Richer variant (RALLY_QUALITY_RESEARCH.md:535) adds a 'Tier' and 'Acceptance / promote criterion' column: | ID | Deliverable | Tier | Depends on | Acceptance / promote criterion | Status |. A measured project also keeps a separate cumulative 'The number' progress table (RALLY_QUALITY_RESEARCH.md:522-531): | after | F1 | merge_rate | ? vs baseline |. Glyph note: template/PROMINENT use ? for done; RALLY_QUALITY_RESEARCH uses ? and its own legend '? not started · ? in progress · ? done' (RALLY_QUALITY_RESEARCH.md:533) ? both accepted.",

"dod_format": "A dedicated '## 9. Definition of done' section (template:64; PROMINENT_COURT_DETECTION.md:112; REQUEST_ACCESS_FORM.md:101). Format = an explicit acceptance checklist of GitHub-style checkboxes '- [] ?', each a verifiable condition (PROMINENT_COURT_DETECTION.md:113-116, REQUEST_ACCESS_FORM.md:102-106). Closing instruction in the template: 'When all ticked ? flip Status ? DONE and archive.' (template:65). For a multi-PR tracked project the DoD may instead be a numbered list of completion clauses tied to tiers/the headline number (RALLY_QUALITY_RESEARCH.md:553-560: 'The project is complete when: 1. ? 2. ? 3. ?'), optionally followed by an owner-adjustable scope note (RALLY_QUALITY_RESEARCH.md:562). Items should be measurable and may carry honesty tags like [measured].",

"style_notes": [

"Numbering: sections are numbered \$0..\$10 starting at '## 0. TL;DR'; §7 is always the progress tracker / source of truth; subsections use dotted numbers (§4.1, §7.1, §7.4) (template:29-69, RALLY_QUALITY_RESEARCH.md:183-560)",

"OPTIONAL sections (§3 Foundation, §5 Threat model) are included only when relevant and DELETED when they don't apply; their slots are reused for domain-appropriate sections (Risks & limits, Experiment plan) (template:14-15,42,48; PROMINENT_COURT_DETECTION.md:77,86)",

```

"Status header is a single multi-line blockquote ('> ') at the very top right under the H1
title, fields joined by ' . ' (template:21-25; PROMINENT_COURT_DETECTION.md:3-7;
REQUEST_ACCESS_FORM.md:3-9)",
"Honesty discipline is mandatory and pervasive: tag every claim inline with backticked
`[verified]` (checked vs code/measured), `[researched]` (needs sign-off), or `[design]`
(proposed); never aspirational ? the doc reflects real shipped state (template:11-12,25;
PROMINENT_COURT_DETECTION.md:7,14,54). Record refuted/killed claims and corrections explicitly
with an 'honesty note' (RALLY_QUALITY_RESEARCH.md:96,364)",
"The doc POINTS to detail (code anchors, research artifacts, sibling docs) and does NOT
duplicate it; $7 is the single source of truth (template:10,13; RALLY_QUALITY_RESEARCH.md:17)",
"Code/config references are concrete and clickable: relative markdown links to sibling
docs e.g. [ `EXPERIMENT_HARNESS.md` ](EXPERIMENT_HARNESS.md), and path:line anchors to code e.g.
backend/pipeline/detectors/tracknet_runner.py:256 (RALLY_QUALITY_RESEARCH.md:79,581;
PROMINENT_COURT_DETECTION.md:57-62)",
"Memory wikilinks [[double-bracket-slug]] cross-reference auto-memory notes, e.g.
[[launch-report-convention]], [[weak-signals-retained]], [[decision-rigor-norm]]
(PROMINENT_COURT_DETECTION.md:7,12,28)",
"Tone: dense, engineering-precise, plainspoken, opinionated; bold for headline takeaways
and the load-bearing word; em-dashes and parentheticals for asides; uses ? for caveats and a
'Dead-ends / do-NOT' list so decisions aren't relitigated (RALLY_QUALITY_RESEARCH.md:261,490;
PROMINENT_COURT_DETECTION.md). Every signal/feature change ships flag-gated, default-OFF,
promoted only on measured lift",
"Per-row workflow rule baked into the legend: 'One small PR per row, branched from
origin/master, no stacking' ? matches CLAUDE.md branching discipline (template:55;
PROMINENT_COURT_DETECTION.md:98; REQUEST_ACCESS_FORM.md:79)",
"Trailing '### Changelog' after a horizontal rule (---), dated bullets newest-appended;
only present once the doc has live history (template:71-74;
PROMINENT_COURT_DETECTION.md:122-126)",
"khelsutra-repo voice (for docs authored there): same status-header + tracker discipline
('mirrors the engine repo's PROJECT_DOC_TEMPLATE'), but tracker may slim to | ID | Deliverable |
Gated? | Status | PR |; privacy-first / contact-only-never-a-training-source guardrail echoed;
portability framed as 'adapter swap, not a rewrite'; comparison tables with two host/option
columns; runbook fenced-code blocks; 'Honest notes' section closing the doc
(REQUEST_ACCESS_FORM.md:6,80; PORTABILITY.md:2,15,44)"
],
"file_refs": [
"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\docs\\PROJECT_DOC_TEMPLATE.md",
"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\RALLY_QUALITY_RESEARCH.md",
"C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\docs\\PROMINENT_COURT_DETECTION.md",
"C:\\Users\\avidu\\Projects\\khelsutra\\docs\\REQUEST_ACCESS_FORM.md",
"C:\\Users\\avidu\\Projects\\khelsutra\\docs\\PORTABILITY.md"
]
}
}

```

Return a concrete, buildable design under this lens. api_additions must reference the REAL gaps found. mvp_scope must be the smallest slice that delivers "pick rallies -> get a reel".

2. assistant (2026-06-20T08:03:28.949Z)

I'll ground this design in the actual research. Let me verify a few key code anchors before finalizing, focusing on the real read path, compile contract, and current static UI.

3. user (2026-06-20T08:03:30.831Z)

```

418         def get_video(self, video_id: str) -> Optional[Dict[str, Any]]:
419             with self._connect() as conn:
420                 row = conn.cursor().execute("SELECT * FROM videos WHERE id = ?",
(video_id,)).fetchone()
421                 if row:
422                     d = dict(row)
423                     d["validation_results"] = json.loads(d["validation_results"] or "[]")
424                     return d

```

```

425         return None
426
427     def query_play_segments(self, video_id: str, filters: Dict[str, Any]) ->
List[Dict[str, Any]]:
428         """
429         Dynamically queries play segments based on filters:
430         - duration_min: float
431         - server: string
432         - receiver: string
433         - event_type: string (e.g. smash, foul)
434         """
435         query = "SELECT r.* FROM play_segments r WHERE r.video_id = ?"
436         params = [video_id]
437
438         if filters.get("duration_min"):
439             query += " AND r.duration >= ?"
440             params.append(filters["duration_min"])
441
442         if filters.get("server"):
443             query += " AND r.server = ?"
444             params.append(filters["server"])
445
446         if filters.get("receiver"):
447             query += " AND r.receiver = ?"
448             params.append(filters["receiver"])
449
450         if filters.get("event_type"):
451             # Check if there is an event of a specific type inside this segment
452             query += " AND EXISTS (SELECT 1 FROM events e WHERE e.segment_id = r.id AND
e.type = ?)"
453             params.append(filters["event_type"])
454
455         segments_list = []
456         with self._connect() as conn:
457             cursor = conn.cursor()
458             rows = cursor.execute(query, params).fetchall()
459
460             for row in rows:
461                 r_dict = dict(row)
462                 r_id = r_dict["id"]
463                 r_dict["metadata"] = json.loads(r_dict["metadata"] or "{}")
464
465                 # Fetch associated events
466                 event_rows = cursor.execute("SELECT * FROM events WHERE segment_id = ?",
(r_id,)).fetchall()
467                 r_dict["events"] = []
468                 for er in event_rows:
469                     e_dict = dict(er)
470                     e_dict["details"] = json.loads(e_dict["details"] or "{}")
471                     r_dict["events"].append(e_dict)
472
473                 segments_list.append(r_dict)
474
475         return segments_list
476
477     def clear_video_data(self, video_id: str):

```

4. user (2026-06-20T08:03:31.344Z)

```

331     @app.post("/api/query")
332     def query_play_segments(req: QueryRequest):
333         filters = {
334             "server": req.server,
335             "receiver": req.receiver,
336             "duration_min": req.duration_min,

```

```

337         "event_type": req.event_type
338     }
339     segments = db_instance.query_play_segments(req.video_id, filters)
340     return {"play_segments": segments}
341
342 @app.post("/api/compile")
343 def compile_highlights(req: CompileRequest):
344     video = db_instance.get_video(req.video_id)
345     if not video:
346         raise HTTPException(status_code=404, detail="Indexed video record not found.")
347
348     # Retrieve matching play segments from db
349     all_segments = db_instance.query_play_segments(req.video_id, {})
350     matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]
351
352     if not matched_segments:
353         raise HTTPException(status_code=400, detail="No matching play segments found to
354 stitch.")
355
356     # Reject path traversal / absolute output names (os.path.join would otherwise let an
357     # absolute or '..'-laden output_filename write anywhere on disk).
358     try:
359         out_name = safe_output_filename(req.output_filename)
360     except ValueError as e:
361         raise HTTPException(status_code=400, detail=str(e)) from e
362
363     # The configured shot-count floor (stitching.min_shot_count) applies on the API
364     # path too ? parity with scripts/run_process_and_stitch.py. Segments WITHOUT
365     shot_count
366     # metadata are exempt: the floor filters known counts only, so explicitly
367     # requested manual/legacy segments can't silently vanish under the default floor.
368     stitching = getattr(global_config, "stitching", None) or StitchingConfig()
369     kept = []
370     for s in matched_segments:
371         meta = s.get("metadata") or {}
372         sc = meta.get("shot_count") if isinstance(meta, dict) else None
373         try:
374             if sc is not None and int(sc) < stitching.min_shot_count:
375                 continue
376             except (TypeError, ValueError):
377                 pass # unparseable count -> treat as unknown, keep
378             kept.append(s)
379         if len(kept) < len(matched_segments):
380             logger.info(f"[compile] stitching.min_shot_count={stitching.min_shot_count}:
381 dropped "
382 f"{len(matched_segments) - len(kept)}/{len(matched_segments)}
383 segments below the floor.")
384         if not kept:
385             raise HTTPException(status_code=400,
386 detail=f"All {len(matched_segments)} matching segments fall
387 below "
388 f"stitching.min_shot_count={stitching.min_shot_count}.")
389
390     # Extract start/end time pairs
391     time_pairs = [(s["start_time"], s["end_time"]) for s in kept]
392
393     # Run compiler with the configured stitching paddings + optional branding watermark
394     output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
395     or "output"
396     compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
397 end_padding=stitching.end_padding,
398 watermark=stitching.watermark)
399     try:
400         out_path = compiler.compile_highlights(video["filepath"], time_pairs, out_name)

```

```

394         return {"status": "success", "filepath": out_path}
395     except Exception as e:
396         raise HTTPException(status_code=500, detail=f"Highlight stitching failed:
{str(e)}") from e
397
398     # --- CLI Debug Utility & Orchestration ---
399     def run_cli_diagnose_clip(video_path: str, timestamp: float):
400         """CLI debugging utility to examine segment properties around a timestamp."""

```

5. assistant (2026-06-20T08:03:34.256Z)

The compile path and read path are confirmed. Now let me verify the current static UI's known bug and the gemini metadata shape to ground the grid card fields.

6. user (2026-06-20T08:03:35.678Z)

```

340
341         if (!res.ok) throw new Error("Failed to filter index.");
342
343         const data = await res.json();
344         renderRallyCards(data.rallies || []);
345     } catch (err) {
346         alert(`Query failed: ${err.message}`);
347     } finally {
348         hideLoader();
349     }
350 };
351
352 // --- 5. COMPILER BOARD MANAGEMENT ---
353 const updateCompilerBar = () => {
354     const count = selectedRallyIds.size;
355     if (count > 0) {
356         compileBoard.style.display = "flex";
357         compileSelectedCount.textContent = `${count} Rally Segment${count > 1 ? 's'
: ''} Selected`;
358     } else {
359         compileBoard.style.display = "none";
360     }
361 };
362
363 btnCompile.addEventListener("click", async () => {
364     if (!currentVideoId || selectedRallyIds.size === 0) {
365         alert("No rallies selected for compiling highlights.");
366         return;
367     }
368
369     const outputName = highlightOutputName.value.trim() ||
"highlights_compiled.mp4";
370     showLoader("Compiling Highlight Reels", "Triggering localized high-speed FFmpeg
stitching demuxers...");
371
372     try {
373         const res = await fetch("/api/compile", {
374             method: "POST",
375             headers: { "Content-Type": "application/json" },
376             body: JSON.stringify({
377                 video_id: currentVideoId,
378                 // API field is segment_ids (the old rally_ids never matched the
request model)
379                 segment_ids: Array.from(selectedRallyIds),
380                 output_filename: outputName
381             })
382         });
383
384         if (!res.ok) {

```

```

385         const err = await res.json();
386         throw new Error(err.detail || "Concatenation failed.");
387     }
388
389     await res.json();
390     // B3 (PR-2): offer the reel via the streaming download endpoint instead of
391     // alerting a server-local path the browser can't reach.
392     const dlUrl =
393     `/api/highlights/${currentVideoId}/${encodeURIComponent(outputName)}`;
394     let dl = document.getElementById("highlight-download-link");
395     if (!dl) {
396         dl = document.createElement("a");
397         dl.id = "highlight-download-link";
398         dl.style.cssText = "display:block;margin-top:10px;font-weight:bold;";
399         compileBoard.appendChild(dl);
400     }
401     dl.href = dlUrl;
402     dl.setAttribute("download", outputName);
403     dl.textContent = `? Download ${outputName}`;
404     alert(`Highlight compilation completed! Use the download link to save
405     ${outputName}.`);
406
407     } catch (error) {
408         alert(`Stitching compilation failed: ${error.message}`);
409     } finally {
410         hideLoader();
411     }
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```

489         if segment_id:
490             segments_data.append({
491                 "id": segment_id,
492                 "start_time": start,
493                 "end_time": end,
494                 "duration": dur,
495                 "metadata": metadata
496             })
497
498     return segments_data
499
500     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> List[Dict[str, Any]]:

```

8. assistant (2026-06-20T08:03:40.722Z)

I have everything confirmed. The known `data.rallies` bug (line 344, API actually returns `play\_segments`), the compile contract (segment_ids, min_shot_count floor exempts unknown counts), the read path (POST /api/query only, no GET listing), and the real per-rally fields (id, start/end/duration, metadata.shot_count + shot_count_confidence + detector). Let me check the khelsutra web/ structure to ground component placement.

9. user (2026-06-20T08:03:42.346Z)

```

web/README.md
---DOCS---
PORTABILITY.md
README.md
REQUEST_ACCESS_FORM.md

```

10. assistant (2026-06-20T08:03:46.492Z)

The khelsutra `web/` is just a README ? the SPA is unbuilt scaffolding, confirming this is a greenfield design over the existing engine API. I have all the grounding I need. The grid-first design must rely on POST /api/query (the only rally-list path) and POST /api/compile + GET /api/highlights download, with honest new-API gaps for listing/thumbnails/clip-preview.

11. user (2026-06-20T08:04:55.049Z)

Structured output provided successfully

12. assistant (2026-06-20T08:05:05.997Z)

I've delivered the grounded design via StructuredOutput. Summary of what I verified against the real engine before answering:

- **Read path** (`backend/main.py:331-340`, `database.py:427-475`): the only rally-listing path is `POST /api/query`, which returns `play\_segments` (id, start_time, end_time, duration, metadata, server/receiver, events). No GET listing, no videos-list endpoint.
- **Compile contract** (`main.py:342-396`): `POST /api/compile {video\_id, segment\_ids, output\_filename}`, applies `stitching.min\_shot\_count` floor (unknown shot_counts exempt), returns a server-local `filepath` not a URL.
- **Download** (`main.py:307-328`): only compiled reels stream via `GET /api/highlights/{video\_id}/{filename}`; no source/clip-span/thumbnail endpoint.
- **Honest fields** (`gemini.py:461-498`): the Gemini path writes only `metadata.shot\_count` + `shot\_count\_confidence` (a string, often "Unknown") + `detector`. `server`/`receiver`/`events` are returned but are random fabrications from the motion segmenter ? the design explicitly does NOT surface them.
- **Known bug** (`app.js:344`): the static UI reads `data.rallies` but the API returns `play\_segments`; the rebuild fixes this.
- **khelsutra `web/`** is currently just `web/README.md` ? confirming this is a greenfield SPA design over the existing API.

The design lands the "pick rallies ? get a reel" loop with **zero** new backend for the MVP

(query + compile + highlights download), and is honest that the visual grid payoff (thumbnails, per-rally preview, a videos-list home, health banner, auth scoping) are all real NEW gaps. Full structured design ? primary surface, data flow, components, the real api_additions, MVP slice, roadmap, conversational integration, pros/cons, and caveats ? is in the StructuredOutput call above.